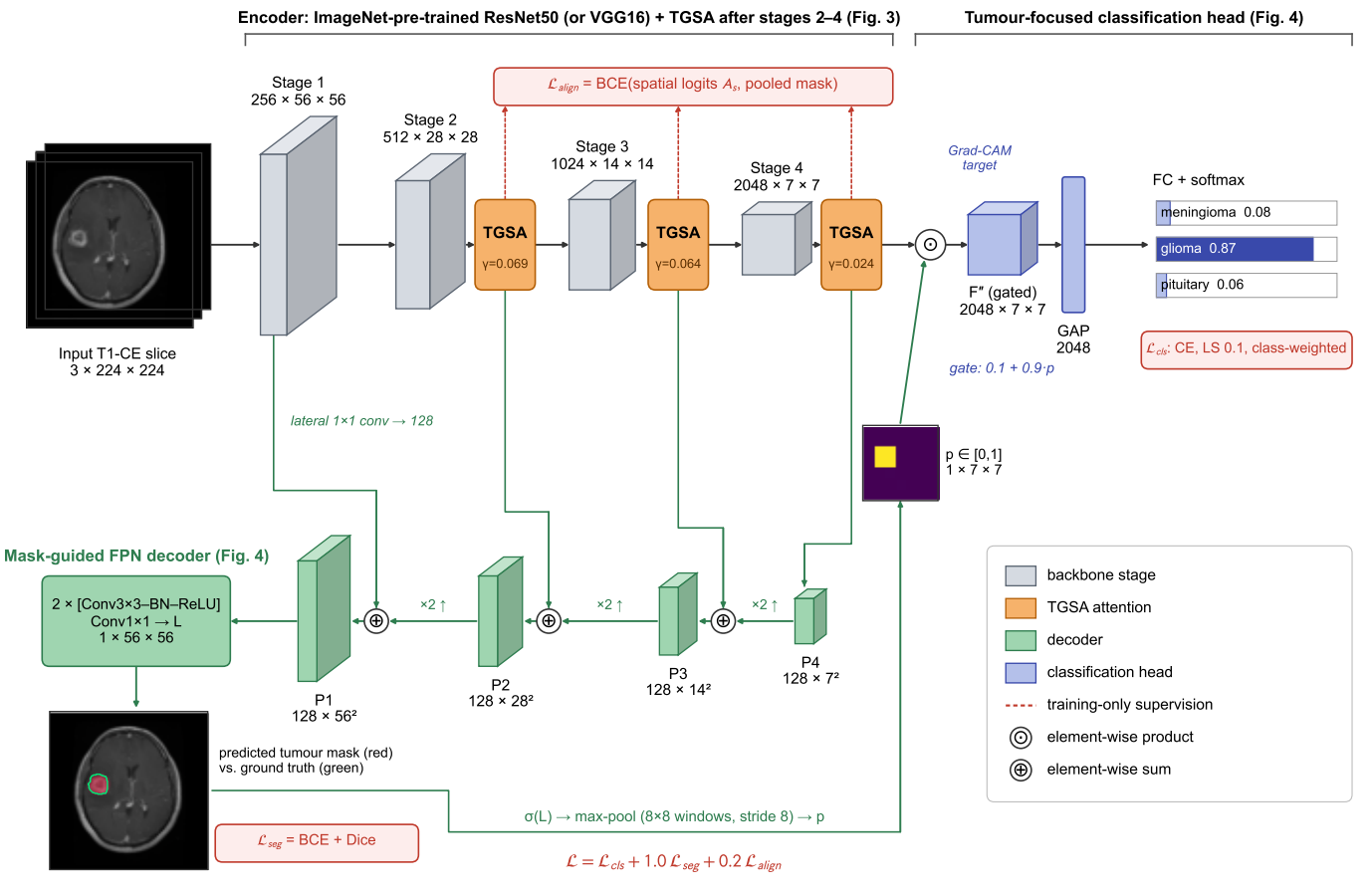

Encoder: ImageNet-pre-trained ResNet50 (or VGG16) + TGSA after stages 2–4 (Fig. 3)
Tumour-focused classification head (Fig. 4)
$\mathcal{L}_{align}$ = BCE(spatial logits A_s, pooled mask)
Stage 1
256 × 56 × 56
Stage 2
512 × 28 × 28
Stage 3
1024 × 14 × 14
Stage 4
2048 × 7 × 7
TGSA
γ=0.069
TGSA
γ=0.064
TGSA
γ=0.024
Grad-CAM target
F″ (gated)
2048 × 7 × 7
GAP
2048
FC + softmax
meningioma 0.08
glioma 0.87
pituitary 0.06
$\mathcal{L}_{cls}$: CE, LS 0.1, class-weighted
gate: 0.1 + 0.9·p
p ∈ [0,1]
1 × 7 × 7
Input T1-CE slice
3 × 224 × 224
lateral 1×1 conv → 128
Mask-guided FPN decoder (Fig. 4)
2 × [Conv3×3–BN–ReLU]
Conv1×1 → L
1 × 56 × 56
×2 ↑
×2 ↑
×2 ↑
P1
128 × 56²
P2
128 × 28²
P3
128 × 14²
P4
128 × 7²
backbone stage
TGSA attention
decoder
classification head
training-only supervision
element-wise product
element-wise sum
predicted tumour mask (red)
vs. ground truth (green)
$\mathcal{L}_{seg}$ = BCE + Dice
σ(L) → max-pool (8×8 windows, stride 8) → p
$\mathcal{L} = \mathcal{L}_{cls} + 1.0\,\mathcal{L}_{seg} + 0.2\,\mathcal{L}_{align}$